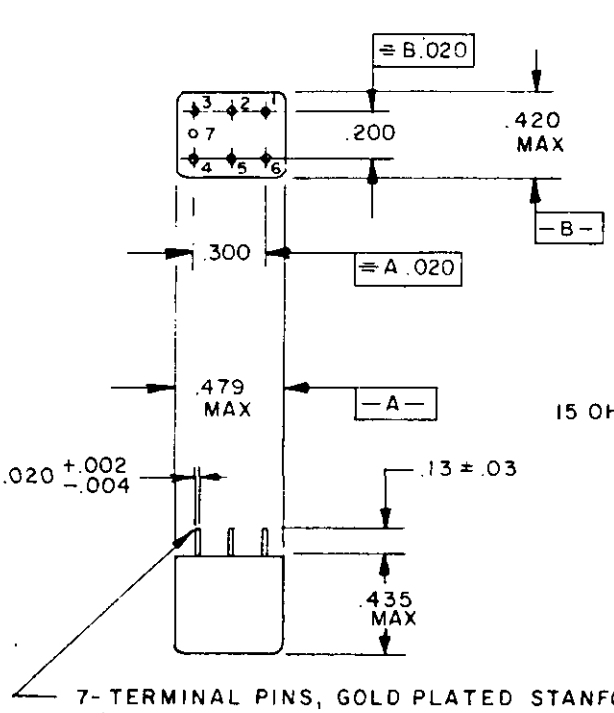
MILITARY SPECIFICATION SHEET

TRANSFORMER, AUDIO FREQUENCY, TF5S21YY944

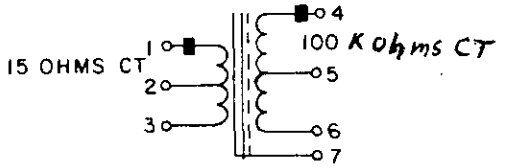
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.



INCHES	MM
.002	.05
.004	.10
.020	.51
.03	.76
.13	3.30
.200	5.08
.300	7.62
.420	10.67
.435	11.05
.479	12.17

POWER LEVEL: .001 MICROWATT



ALTITUDE - 10,000 FT MAX
WORKING VOLTAGE 25 V MAX

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Marking shall be on the sides and top of the case.

FIGURE 1. Dimensions and configuration.

FSC 5950

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-3) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:
Source impedance (1-3): 15 ohms maximum.
Load impedance (4-6): 100 kilohms in parallel with a 20 picofarads (pF) capacitor.
Primary power level: .001 microwatt.
Primary voltage (1-2-3): 0.5 millivolt, 15.6 kHz ±500 Hz sine wave.
Secondary voltage (4-5-6): 10 millivolts.
Working voltage: 25 volts.
DC resistance: (1-3) 3.4 ohms maximum.
(4-6) 740 ohms maximum.

Design and construction:
Dimensions and configuration: See figure.
Duty cycle: Continuous.
Altitude: 10,000 feet.
Case: Metal, epoxy filled.
Terminal type: Pin, 0.020 diameter, gold-plated, printed circuit, Stanford Applied Eng. or equal.
Weight: 4 grams maximum.
Operating temperature range: -55° to $+130^{\circ}$C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage (at sea level): 20 Vrms.

Electrical characteristics:
No load (center-tap voltage unbalance):
(a) With 0.2 V rms, 20 kHz, 0 ampere dc applied to (1-3), voltage across (4-6) shall be 4 volts ±1 percent.
(b) With 0.2 V rms, 15.75 kHz, 0 ampere dc applied to (1-3), current in (1-3) shall be 0.67 mA maximum.
Harmonic distortion: 3 percent maximum at .001 microwatt from 10 kHz to 20 kHz.
Primary impedance: 148 ohms ±10 percent with 0.2 V rms, 15.6 kHz applied to (1-3) and (4-6) loaded with 100 kilohms in parallel with 20 pF capacitor.
Secondary load impedance: 100 kilohms in parallel with a 20 pF capacitor.
Electrostatic shielding: The voltage ratio shall be a minimum of 3 to 1 at 20 kHz.
Self-resonant frequency: 200 kHz minimum with (4-6) shorted.
Frequency response: 10 kHz to 20 kHz, ±3 dB, reference frequency 15.6 kHz.
Phase shift: Primary (1-3) to secondary (4-6) with secondary (4-6) loaded with 100 kilohm resistor and 20 pF capacitor in parallel and 0.2 V rms, 15.6 kHz applied to (1-3). Limit: 0° $\pm 3^{\circ}$.
Polarity: Additive, with terminals 3 and 4 connected.

Shock (specified pulse): MIL-STD-202, method 213, test condition K, under no load condition.

Marking location: See figure.
Part number: M27/125-01.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, SG
Navy - EC, SH, OS
Air Force - 11, 17, 99
DSA - ES

User activities:
Army - MU, WC, ME
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5950-0522-9)

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